llmXive Automated Review of DataComp-VLM: Improved Open Datasets for Vision-Language Models

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper presents a comprehensive study on data curation for Vision-Language Models, with a strong focus on decontamination and data mixing. The claims regarding the decontamination pipeline (Section 5) are well-supported by the detailed methodology and the removal rate statistics provided in the text and figures. The claim that "filtering rarely helps" (Section 6) is supported by the extensive ablation studies in the appendices.

However, there are a few specific instances where the textual claims appear to contradict the visual data or require clarification to ensure the reader can trust the numbers:

1. Section 6.1 (Data Mixing): The text states, "the Instruction-heavy mix is the worst at 1Bx6.25B". However, visual inspection of Figure 4 (mixing-panels) for the 1B model at 6.25B tokens suggests the red line (Instruction-heavy) is actually the highest (best) or at least not the worst compared to the other lines. This is a direct contradiction between the text and the figure. The authors must verify the data plotted in Figure 4 and correct the text to accurately reflect the results.
2. Section 3.1 (Pool Composition): The text claims the pool has 3.9B samples and 6.0T tokens. The caption of Figure 1 states image-caption pairs are 83% of tokens. While the text lists the number of datasets, it does not explicitly break down the 3.9B/6.0T totals by data type to allow the reader to verify the 83% claim. A brief breakdown or a reference to a table with these specific totals would strengthen the claim.
3. Section 5.1 (Decontamination Rates): The text states that "only 0.29% of all training samples are removed" at the pool level, despite high removal rates in specific datasets like InfoVQA (100%) and ScienceQA (66.4%). Given the total pool size (3.9B), this 0.29% figure implies a very small absolute number of removed samples. The text should explicitly confirm that this 0.29% is the global average across the *entire* 3.9B sample pool to avoid confusion about the impact of the decontamination.
4. Section 6.2 (SFT Correlation): The claim of a Pearson correlation of 0.99 is extremely high. The text mentions 27 checkpoints and 54 SFT runs. It is crucial to clarify that the correlation is calculated on the 27 pretraining scores vs. the 27 corresponding post-SFT scores (likely averaged), not on the 54 individual runs. The figure caption should also explicitly state the number of points used for the correlation calculation.

These issues are primarily about ensuring the text and figures are perfectly aligned and that the reader can verify the specific numbers. They do not invalidate the core findings but require correction for accuracy.

Action items.

- [**writing**] Section 3.1 claims 3.9B samples/6.0T tokens, but the 83% token share for image-caption pairs isn't verified by the text breakdown. Please add a table or sentence confirming the 3.9B/6.0T totals sum correctly with the component counts to support the 83% claim.
- [**writing**] Section 4.1 states 'English data carries almost all signal (0.6pp drop if removed)' based on Table 2 (All 49.1 vs Eng-only 48.5). However, the text also says 'Non-English tail

alone is insufficient (-3.0pp)'. Clarify that the 0.6pp drop refers to removing non-English data, not English, to match the table comparison and avoid confusion.

- **[writing]** Section 5.1 claims 'only 0.29% of all training samples are removed' globally despite high rates in specific datasets (e.g., InfoVQA 100%). Confirm this 0.29% is the global average across the entire 3.9B pool, not just the subset in Figure 3, to validate the 'cheap' decontamination claim.
- **[science]** Section 6.1 states the 'Instruction-heavy mix is the worst at 1Bx6.25B', but Figure 4 shows the red line (Instruction-heavy) as the highest (best) at that point. Verify the data in Figure 4 and correct the text or figure labels to resolve this direct contradiction.
- **[writing]** Section 6.2 claims Pearson r=0.99 between pre- and post-SFT scores for 27 checkpoints. Clarify if this correlation is calculated on the 27 checkpoint averages (not 54 runs) and ensure the figure caption explicitly states the number of points used.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 is a clear and well-structured flowchart that effectively visualizes the data collection pipeline described in its caption. All steps, data types, and final corpus sizes are legible and logically connected.

Figure 2

The figure effectively visualizes the scaling trends described in the caption, but the legend abbreviations ('IC', 'IT') contradict the caption's notation ('c', 'i'), and the y-axis unit label is slightly ambiguous.

Figure 3

Figure 3 effectively displays the line search results with clear annotations for best points, but the caption's claim of 'competitive' baseline at small scale lacks quantitative support, and the differing y-axis scales between subplots hinder direct comparison.

Figure 4

The figure effectively visualizes the filter ablation results with clear color coding and value labels. However, the title text contains a minor contradiction regarding the magnitude of gains (‘< 1%’ vs ‘>= +0.5%’) and includes redundant information that clutters the header.

Figure 5

Figure 5 is clear and effectively supports its caption. The bar chart clearly displays the ‘No Filter’ vs. ‘NVIDIA Mixtral Text-Only’ comparison across four mixture/model configurations, with explicit numerical labels and delta annotations (Δ) that match the caption’s claims.

Figure 6

The figure clearly compares sample-level versus shard-level sampling across four configurations on validation and core sets. The legend, axis labels, and units are present and legible, and the green annotations accurately reflect the performance gains described in the caption.

Figure 7

The figure effectively visualizes the performance drop at extreme temperatures, but the caption contains a typo regarding the flattening condition, and the orange baseline line implies a continuous trend without data points.

Figure 8

The figure clearly displays the comparison data with appropriate labels and values, but the caption’s claim that the method outperforms in 5 of 6 categories is factually incorrect based on the visual data, which shows it trailing in two categories (Knowledge and OCR).

Figure 9

The figure is a clear and well-organized bar chart that effectively visualizes per-dataset score changes. The caption accurately describes the data trends and the average drop line, though it contains a minor grammatical omission and slightly simplifies the dataset names compared to the axis labels.

Figure 10

The bar chart effectively visualizes the performance gains from upsampling OCR data, but the figure caption contains a grammatical error where the sentence trails off after ‘decontaminated’, failing to name the baseline model.

Figure 11

The figure effectively demonstrates the negligible overhead of online filtering as claimed in the caption, but the rightmost bar chart is missing a y-axis label to explicitly define the plotted metric.

Figure 12

The figure is visually clear and the legend is well-defined, but the x-axis label contains a typo (‘orignal’). Additionally, the title’s claim of ‘no significant gain’ is visually contradicted by the +1.2% data point in the MTL category, which lacks error bars to statistically justify the dismissal.

Action items.

- [**writing**] Figure 2: The legend uses ‘IC’ and ‘IT’ (e.g., ‘10% IC, 70% IT’), but the caption defines the mixtures using ‘c’ and ‘i’ (e.g., ‘10c-70i’). This inconsistency between the visual legend and the text description is confusing.
- [**writing**] Figure 2: The y-axis label ‘Micro Avg (%)’ is ambiguous; it is unclear if the values (e.g., 41, 42) represent raw percentages (41%) or a scaled score (e.g., 41.0 out of 100).
- [**science**] Figure 3: The caption claims the optimal configuration at medium scale is ‘70%IT (+2.0% over baseline)’, but the chart shows the baseline (dotted line) at ~54.3% and the best point at 56.3%, which is a +2.0% absolute difference, not a relative one. However, the text says ‘+2.0% over baseline’ which is ambiguous (could mean percentage points or relative percent). Given the context of micro avg %, it likely means percentage points, but the phrasing is slightly imprecise. More critically, the capti
- [**writing**] Figure 3: The y-axis label ‘Micro Avg (%)’ is present, but the tick labels on the left plot start at 41.0 and go to 43.5, while the right plot starts at 54.25 and goes to 56.25. This makes direct visual comparison between scales difficult. Consider using a shared y-axis scale or adding a note explaining the different baselines.
- [**writing**] Figure 4: The title text ‘Only 3 of 39 filters show any gain — all < 1% and wash out at larger scale’ contradicts the caption’s claim that gains are ‘>= +0.5%’; the title’s ‘< 1%’ is ambiguous and potentially misleading regarding the magnitude of the positive results.
- [**writing**] Figure 4: The title text ‘Small scale (1B model, 6.25B tokens)’ is redundant with the caption and makes the header cluttered; this information is better placed in the caption or a subtitle.
- [**writing**] Figure 7: The caption contains a typo ‘flattening ($T2$)’ which is missing the inequality symbol (likely $T \geq 2$ or $T > 1$) and does not match the x-axis label ‘2.0 (sqrt)’.
- [**science**] Figure 7: The orange line labeled ‘T=1 baseline’ extends from T=1.0 to approximately T=2.5 without data points, implying a trend that is not supported by the discrete sampling shown for the red line.
- [**science**] Figure 8: The caption claims a +13.2% gain in ‘General’ and +18.1% in ‘Vision’,

but the chart shows +13.2% (68.4 vs 55.2) and +18.1% (57.2 vs 39.1) respectively. However, the caption states 'outperforms on 5 of 6 categories', but the chart shows 'Ours' trailing in 'Knowledge' (67.6 vs 68.0) and 'OCR' (54.1 vs 58.9). This is only 4 out of 6 categories where 'Ours' outperforms, contradicting the caption's claim of 5.

- [**writing**] Figure 8: The title mentions '+5.8% Micro Avg improvement', which matches the chart (58.9 vs 53.1), but the caption does not mention this specific value, creating a disconnect between the visual title and the textual description.
- [**writing**] Figure 9: The caption states 'The largest drops are on ScienceQA (-10.7%), nq (-7.6%), and ChartQA (-5.9%)', but the chart shows 'ChartQA_TEST' at -5.9% and 'nq' at -7.6%, while 'ScienceQA_VAL' is -10.7%. The caption omits the '_VAL' and '_TEST' suffixes present in the chart labels, creating a minor inconsistency in dataset naming.
- [**writing**] Figure 9: The caption text '...categories where appeared strongest' contains a missing subject (likely 'DataComp-VLM' or 'the model'), making the sentence grammatically incomplete.
- [**writing**] Figure 10: The caption ends with 'closing roughly half the remaining gap with decontaminated .' which is grammatically incomplete and missing the noun (likely 'FineVision') that the bar labels imply.
- [**science**] Figure 10: The title claims 'Upsampling OCR to 25% yields +3.1% gain', but the visual comparison between the 'Ours (standard)' bar (45.8) and the '+ OCR Upsampled 25%' bar (48.9) shows a difference of 3.1, which is correct, yet the caption text 'closing roughly half the remaining gap' is vague without explicitly stating the gap size to the FineVision baseline in the text.
- [**writing**] Figure 11: The rightmost bar chart lacks a y-axis label; the axis shows values (0.90–1.15) but does not explicitly state the unit or metric (e.g., 'Runtime multiplier' or 'Relative runtime'), relying solely on the caption for context.
- [**writing**] Figure 12: The x-axis label contains a typo ('orignal' instead of 'original').
- [**science**] Figure 12: The title claims 'no significant gain,' but the 'MTL' category shows a +1.2% gain for synthetic spatial captions, which appears to contradict the title's absolute claim without a statistical significance indicator (e.g., error bars) to justify the dismissal.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-structured for a technical audience, but it relies on several subfield-specific acronyms and named methods that are not defined at their first occurrence. A competent

PhD from an adjacent field (e.g., NLP or general ML) might stumble on terms like "Q-formers," "SSCD," and the abbreviation "pp" (percentage points), as these are not universal standard vocabulary across the broader discipline.

Specifically, "Q-formers" appears in the Related Work without explanation, assuming the reader knows the specific architecture of BLIP-2. Similarly, "SSCD" is introduced as a proper noun for a specific descriptor model without expanding the acronym or briefly explaining its function beyond "copy-detection." The use of "pp" for percentage points is a common source of confusion for non-specialists who might interpret it as a relative percent change. Finally, "MTL" is used as a category label for benchmarks; while likely "Multilingual" here, the ambiguity with "Multi-Task Learning" creates a potential barrier to immediate comprehension.

These are minor, easily fixable issues that do not detract from the paper's quality but are necessary to ensure the work is accessible to the "adjacent-field PhD" standard required by the review panel. Defining these terms at first use would eliminate the need for the reader to guess or search external sources.

Action items.

- [**writing**] Section 'Extended Related Work' (Appendix) uses 'Q-formers' without definition. While 'cross-attention' is mentioned, the specific architecture component 'Q-former' (Query-former) is a named method from BLIP-2 that may not be known to all adjacent-field readers. Add a brief gloss: 'Q-formers (query-based cross-attention modules)'.
- [**writing**] Section 'Train-Test Decontamination' (Appendix) introduces 'SSCD' (self-supervised copy-detection descriptor) and 'MinHash' without expansion. While 'MinHash' is standard in NLP, 'SSCD' is specific to this pipeline. Define SSCD at first use: 'SSCD (self-supervised copy-detection descriptor)'.
- [**writing**] Section 'Train-Test Decontamination' (Appendix) uses 'Jaccard' similarity without defining the metric or the set operation (intersection over union) for a reader who might know MinHash but not the specific similarity metric used. Add: 'Jaccard similarity (intersection over union of the n-gram sets)'.
- [**writing**] Section 'Evaluation Suite Details' (Appendix) lists 'MTL' as a benchmark category without defining the acronym. In this context, it likely means 'Multilingual', but 'MTL' often stands for 'Multi-Task Learning' in adjacent fields. Define explicitly: 'MTL (Multilingual)'.
- [**writing**] Section 'Data Mixing' (Appendix) uses 'pp' as an abbreviation for 'percentage points' (e.g., '+1.1pp'). While common in economics and some ML subfields, it is often confused with 'percent' by readers from adjacent disciplines. Define at first use: 'percentage points (pp)'.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper's argument structure is logically sound and internally consistent. The central thesis—that data mixing strategies are scale-dependent and that quality filtering yields diminishing returns at scale—is supported by a coherent chain of reasoning: the authors establish a baseline, introduce controlled variables (mixtures, filters), and demonstrate that outcomes reverse or stabilize depending on model scale and token budget.

Specifically, the transition from the "Filtering rarely helps" section to the "Data Mixing" section is well-motivated. The authors correctly identify that the lack of filtering gains (Section 4) suggests the underlying mixture distribution is the primary driver of performance, leading logically to the mixing experiments in Section 5. The conclusion that "Data mixing cannot be scale agnostic" follows directly from the empirical observation in Figure 4 (and Table 2 in the appendix) where the "Instruction-heavy" mix performs worst at small scales but best at large scales. This is a valid inductive inference from the presented data.

The decontamination protocol (Appendix) is defined with precise thresholds (SSCD ≥ 0.75, MinHash ≥ 0.55) and the rationale for these choices (balancing precision/recall on non-natural images) is consistent with the reported removal rates. There are no contradictions between the stated methodology and the reported results (e.g., the removal rates in Figure 5 match the textual description).

Definitions of data types (image-caption, instruction-tuning, etc.) remain stable throughout the text and tables. The distinction between "Core," "Extended," and "Validation" suites is clearly defined and applied consistently in the results tables. No non-sequiturs or circular arguments were detected. The argument holds together as a valid proof of the paper's claims given the stated premises.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims about the generalizability of its findings regarding data filtering and mixing that exceed the scope of the provided evidence.

First, the abstract and conclusion assert that the study demonstrates "no individual quality filter provides reliable and consistent gains," effectively suggesting that filtering is obsolete for VLM training. However, the data in Appendix E003 (Tables for local filtering) shows that while many filters fail, specific configurations (e.g., NVIDIA Nemotron on text-only data at medium scale, or OpenAI-CLIP on image-caption pairs) do yield small but positive gains (+0.4pp to +0.6pp) over the no-filter baseline. The rhetoric of "no reliable gains" ignores these positive signals and overgeneralizes the null results found in other configurations. The claim should be hedged to reflect that filtering is *inconsistent* rather than universally ineffective, or the authors should demonstrate that these small gains are statistically insignificant or not robust across seeds.

Second, the paper frames its findings on data mixing (specifically the superiority of "Instruction-heavy" mixtures at larger scales) as a universal principle for VLMs. The experiments, however, are conducted exclusively on the Qwen2.5 family of models (1B, 2B, 4B) using a specific InternVL-style training recipe. The conclusion that "scale-aware curation is required" and that instruction-heavy mixes are superior is presented as a general law, yet it is only demonstrated for one model family and one architecture. Without testing on a structurally different model (e.g., a Q-former based model like Flamingo or a different LLM backbone), the claim of universality is an overreach. The text should explicitly limit the scope of the generalization to the tested model families or include a cross-architecture validation.

Finally, the paper uses strong language like "solves" or "eliminates" regarding the problem of data contamination or the necessity of filtering. While the decontamination section is rigorous, the claim that the proposed method "solves" the problem of train-test overlap is too absolute given that the method relies on specific thresholds (SSCD 0.75, MinHash 0.55) which are heuristic choices. The paper admits these thresholds involve trade-offs (false positives/negatives). The rhetoric should be adjusted to "mitigates" or "significantly reduces" rather than "solves," and the limitations of the threshold selection should be more prominently acknowledged in the conclusion rather than buried in the appendix.

Action items.

- [**writing**] The abstract/conclusion claims the method 'solves' or 'eliminates' the need for quality filtering, but Section 4 and Appendix E002 show filtering provides diminishing returns or small gains only on specific subsets (e.g., global CLIP-score on small scale). Replace absolute terms like 'solves' with 'reduces the efficacy of' or 'shows diminishing returns for' and scope the claim to the tested model scales (1B-4B) and token budgets.
- [**writing**] The paper states 'no individual quality filter provides reliable and consistent gains' (Section 2/Abstract), yet Table E003 (medium scale) shows NVIDIA Nemotron (text-only) and OpenAI-CLIP (im-cap) achieving +0.4pp to +0.6pp gains over the baseline in specific categories. The claim of 'no reliable gains' overgeneralizes the mixed results. Qualify the claim to 'no consistent gains across all data types and scales' or explicitly acknowledge the specific conditions where filters did help.
- [**science**] The conclusion implies the findings generalize to 'all VLMs' or 'any scale,' but the experiments (Section 4, Appendix E001) are restricted to Qwen2.5-based models (1B, 2B, 4B) and a specific training recipe (InternVL-style). The claim of universality is not licensed by the single-family study. Narrow the claim to 'within the Qwen2.5 family and tested scales' or add experiments on a distinct architecture (e.g., LLaVA or Flamingo-style) to support broader generalization.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a large-scale dataset curation and training study for Vision-Language Models (VLMs). From a safety and ethics perspective, the work is low-risk and well-documented.

The authors explicitly address the primary ethical concern for data-centric ML papers: **data provenance and licensing**. Section `appsub:licensing` and the extensive license table (spanning `e003`) meticulously list the source, license type (e.g., CC-BY, Apache-2.0, MIT, or "Unknown" where applicable), and access links for all 160 datasets in the pool. This transparency allows downstream users to verify compliance with their own usage constraints.

Furthermore, the paper demonstrates rigorous **decontamination practices** (Section `app:decontamination`) to prevent train-test leakage, which is a standard integrity requirement rather than a safety risk, but the detailed methodology (SSCD for images, MinHash for text) shows a commitment to scientific rigor. The exclusion of grounding benchmarks (RefCOCO variants) due to contamination risks further underscores this careful approach.

There are no indications of:

- **Human-subjects data** requiring IRB approval (the data consists of public web crawls, synthetic data, and standard academic benchmarks).
- **PII exposure** (the paper does not release raw scraped content with personal identifiers; it aggregates and filters).
- **Dual-use capabilities** that lower the barrier to specific harms (the work improves general VLM performance, a standard capability in the field, without introducing novel offensive or deceptive mechanisms).
- **Undisclosed conflicts of interest** (the authors are from a mix of academic and industry labs, but the work is presented as open research with open data/code links).

The paper adheres to the norms of the field regarding dataset construction and does not present foreseeable, non-trivial risks that are unaddressed. No action items are required.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a rigorous investigation into VLM data curation, particularly regarding decontamination and mixing strategies. However, the evidentiary strength of the central claims regarding data mixing and filtering is weakened by a lack of variance reporting and potential confounds in experimental design.

First, the headline findings on data mixing (Section 4.2, Figure 3) and learning rate selection (Appendix, Table 1) are presented as single-run results. The reported improvements, such as the shift from "worst" to "best" performance for the Instruction-heavy mix at larger scales, are often in the range of 0.5% to 2.0%. In deep learning training, such margins are frequently indistinguishable from the variance introduced by random weight initialization or data shuffling. Without reporting results across multiple seeds (e.g., mean ± standard deviation), the reader cannot determine if these "scale-aware" effects are robust phenomena or lucky seeds. This is critical for a paper claiming to establish generalizable data mixing laws.

Second, the data mixing experiments (Section 4.2) suffer from a potential confound regarding compute allocation. When the authors shift from a "Caption-heavy" to an "Instruction-heavy" mixture, they change the proportion of data types. However, the total number of tokens processed per epoch or the total training budget per data type is not explicitly controlled to be identical across the mixture variants in a way that isolates the *ratio* effect from the *exposure* effect. If the "Instruction-heavy" model simply sees more instruction tokens (even if the total token budget is fixed, the *density* changes), the gain might be due to increased exposure to that specific data type rather than the optimal *mixing ratio* itself. A control run that matches the exact token count of the baseline for the non-instruction data types (via repetition or subsampling) is needed to isolate the mixing ratio as the causal factor.

Finally, the claim that "filtering rarely helps" (Section 4.1) compares results across different evaluation suites: the small-scale experiments use the 13-task Validation suite, while the medium-scale experiments use the 33-task Core suite. This inconsistency introduces a confound where the observed "diminishing returns" of filtering could be an artifact of the different benchmark distributions (e.g., the Validation suite might be more sensitive to noise reduction than the Core suite) rather than a true scaling law. To robustly support the claim that filtering effects vanish at scale, the authors must report filtering results on a consistent, fixed evaluation suite across all model scales.

Addressing these design gaps—specifically by adding seed variance, controlling for token exposure in mixing experiments, and standardizing the evaluation suite—would significantly strengthen the causal claims regarding data curation strategies.

Action items.

- [**writing**] The paper presents a rigorous investigation into VLM data curation, particularly regarding decontamination and mixing strategies. However, the evidentiary strength of the central claims regarding data mixing and filtering is weakened by a lack of variance reporting and potential confounds in experimental design. First, the headline findings on data mixing (Section 4.2, Figure 3) and learning rate selection (Appendix, Table 1) are presented as single-run results. The reported improvements, such a

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical treatment in this paper generally aligns with ML field norms regarding the omission of formal hypothesis tests for benchmark comparisons. However, there is a consistent lack of uncertainty reporting for primary quantitative claims, which undermines the reliability of the reported improvements.

In **Sections 4.2 and 5**, benchmark scores (e.g., 68.4, 73.0) are presented as single point estimates. The manuscript does not specify the number of training seeds or provide standard deviations (SD), standard errors (SE), or confidence intervals (CI). In deep learning, performance varies across random seeds. Reporting a single number creates a false sense of precision. A reported gain of 0.6 percentage points is statistically indistinguishable from noise if the standard deviation across seeds is >0.5. The authors should report mean ± SD over at least 3 seeds for all main results or explicitly state that numbers come from a single run.

In **Section 3.2**, the authors report a Pearson correlation of $r = 0.99$ and Spearman $\rho = 0.99$ based on 27 data points. While the correlation is strong, the statistical significance (p-value) and 95% confidence interval are missing. With $N = 27$, omitting the confidence interval leaves the inferential claim incomplete.

Finally, in **Sections 3.1 and 4.1**, specific point-difference improvements (e.g., "+1.1pp") are reported to one decimal place. Without accompanying variance metrics, these specific decimal claims are not statistically justified. If variance is not reported, the precision of the reported differences should be reduced (e.g., to integers) to avoid false precision.

These are reporting gaps rather than fundamental errors in test selection, but they prevent the numbers from meaningfully supporting the paper's claims of robust improvements.

Action items.

- [**writing**] Sections 4.2 and 5 report benchmark scores as single point estimates without uncertainty (SD/SE/CI). Report mean ± SD over 3 seeds for main results, or explicitly state results are from a single run to avoid implying false precision.
- [**writing**] Section 3.2 reports Pearson r=0.99 and Spearman =0.99 for N=27 without confidence intervals or p-values. Add 95% CIs for these correlation coefficients to support the statistical significance of the claim.
- [**writing**] Sections 3.1 and 4.1 report specific decimal improvements (e.g., +0.6pp) without reported variance. Without SD across seeds, these decimal claims are unsupported. Round differences to integers or report variance to justify the precision.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript "DataComp-VLM: Improved Open Datasets for Vision-Language Models" demonstrates a high standard of writing quality. The prose is clear, concise, and allows the reader to move through the complex technical details without friction.

The structure is logical and well-signposted. The paper begins with a clear statement of contributions, followed by a detailed exposition of the data pool construction, decontamination protocols, and experimental results. Each section flows naturally into the next, with effective transitions that orient the reader to the purpose of the upcoming content. For instance, the transition from the general dataset description to the specific decontamination methodology is smooth and motivated by the need for rigorous evaluation.

Paragraphs are well-constructed, each focusing on a single main point. Topic sentences are generally clear and appear at the beginning of paragraphs, allowing the reader to quickly grasp the paragraph's intent. The authors avoid burying key claims in the middle or end of paragraphs. Sentences are grammatically correct and free of garden-path constructions or ambiguous pronoun references. The tense and voice are consistent throughout the document, maintaining a professional and objective tone.

The abstract effectively summarizes the paper's method, results, and conclusions, providing a clear roadmap for the reader. Terminology is used consistently; for example, "DCVLM" and "pool" are used uniformly to refer to the dataset and its components. The figures and tables are well-integrated into the text, with captions that are descriptive and easy to parse.

While the paper is dense with information, the writing manages this density effectively. The use of bullet points and tables helps to break up complex data and makes it more digestible. The authors also do a good job of explaining technical terms and concepts, ensuring that the paper is accessible to a broad audience within the field.

In summary, the writing quality of this paper is excellent. It is well-organized, clearly written, and free of the common readability issues that can hinder comprehension. No revisions are necessary from a writing quality perspective.